Mitsue Forest → Energy: A Workforce-Led 25-Year Plan

> **"The forest our ancestors planted — the power that sustains the village they built"** 先人が植えた森が、今、村を支える力になる。

1. The core idea

The bottleneck in Mitsue's forest is **not the trees — it is the crew and the machines.** 御杖村森林組合 (Mitsue Kanko) today manages only a fraction of the village's **7,051 ha** of forest, most of it as subsidised thinning rather than full harvest.

This project's purpose is to **double — and over time triple — the cooperative's workforce and mechanise it** (harvester + forwarder, chipper, drying). Energy revenue from the harvested sugi pays for that crew and equipment; the larger crew harvests more; more harvest feeds a larger biomass plant. It is a **self-funding loop**, the same model proven in Shimokawa and Nishiawakura.

2. Where the cooperative is today

Item	Current	Source
Village forest area	~7,051 ha (>90% private, sugi/hinoki postwar plantations)	Co-op report
Annual thinning	~66 ha/yr	Co-op report
Raw material recovered	only ~120–300 m³/yr	Co-op report
Crew	small, young (avg age ~33)	Co-op report

The recovered volume is tiny relative to the area worked because the current regime is **subsidised thinning, not mechanised harvest**. That is precisely the gap the project closes.

3. Real precedents we can check

Village / Town	Forest	Harvest model	Energy result
Shimokawa, Hokkaido (下川町)	4,500+ ha town forest (FSC 7,150 ha)	**~50 ha/yr** harvest **+ replant**, 60-yr sustained cycle	56% local heat self-sufficiency; −20% town CO_2
Nishiawakura, Okayama (西粟倉村)	93–95% forest	~**3,000 ha** under 100-yr management contracts (village manages on owners' behalf)	forestry-jobs revival
Maniwa, Okayama (真庭市)	city-scale	**148,000 t/yr** (90k unused wood + 58k mill residue)	**10 MW**, 79.2 GWh/yr, ~22,000 homes, 95% uptime
Mishima, Fukushima (三島町)	~88% forest	~750 t/yr	≤50 kWe — our model's calibration anchor

Shimokawa is the closest sustained-cycle model: a whole town forest run on **~50 ha/yr by a professional, mechanised crew**. It shows what a scaled-up Mitsue Kanko can look like.

4. What the workforce can realistically deliver

Harvest volume — and therefore plant size — is set by how large and how mechanised the crew becomes. Figures below are from the project's calibrated **Forest Twin** model (sugi 600 m³/ha, 314 kg/m³, 18.5 GJ/t; net electrical efficiency 0.13, calibrated so a Mishima-scale unit reproduces ~50 kWe on ~750 t/yr). All harvested wood is routed to fuel; conversion runs over 25 years.

Cooperative capacity	Harvest rate	Area in 25 yr	Wood / yr	Biomass plant	Electricity	Heat
Current + light mechanisation	~70 ha/yr	1,762 ha (25%)	10,500 m³	~290 kWe	~2,000 MWh/yr	~6,900 MWh-th/yr
Double crew + harvester/forwarder ✅	**~140 ha/yr**	**3,526 ha (50%)**	42,000 m³	**~1,150 kWe**	**~8,000 MWh/yr**	~27,700 MWh-th/yr
Triple crew, full mechanisation	~200 ha/yr	~5,000 ha (~70%)	~60,000 m³	~1.6 MWe	~11,000 MWh/yr	~38,000 MWh-th/yr
(Optimistic 90% — needs ~4× crew)	*254 ha/yr*	*6,346 ha (90%)*	*135,000 m³*	*3.7 MWe*	*~26,000 MWh/yr*	*~89,000 MWh-th/yr*

✅ **Recommended baseline: the doubled-workforce case — ~50% of the forest, ~1.1–1.2 MWe.**

5. Recommended plant

Build the plant as **2 × ~0.6 MWe units**, not one 1.2 MWe unit:

- **Redundancy / maintenance** — one unit can be serviced while the other runs.
- **FIT** — both stay under the 2 MW unused-material (未利用材) band, capturing the higher tariff (~¥40/kWh vs ~¥32/kWh for ≥2 MW).
- **Phased growth** — install unit 1 as the crew doubles, unit 2 as it approaches triple. The plant grows in step with the workforce.

Indicative capex at this scale ≈ ¥0.8 billion (¥700k/kWe installed). Electricity → grid or salt-battery thermal store; the recovered heat (~27,700 MWh-th/yr) also charges the store or supplies district heat.

Reference equipment — makers

Small woody-gasification units mostly top out at ~200–500 kWe, so ~0.6 MWe is a **cluster of modules** either way. Prefer **domestic makers** where the size fits — faster service/parts, simpler FIT + 交付金 paperwork, and it reinforces the local supply-chain / endogenous-development story.

Maker	Origin	Tech / scale	Notes
中外炉工業 (Chugai Ro)	[J][P]	rotary-kiln gasification cogen	Since 1997, NEDO-backed; tolerates rough fuel (branches/bark)
神鋼環境ソリューション (Shinko / Kobe Steel)	[J][P]	~500 kWe-class up	Confirm gasification vs steam (their big plants are combustion)

Maker	Origin	Tech / scale	Notes
静岡製機 (Shizuoka Seiki)	[J][P]	small ~200 kWe outdoor unit	METI-funded, aimed at agriculture/forestry — closest in spirit
ネオナイト (Neonite)	[J][P]	downdraft two-stage	Recovers charcoal + wood tar (biochar angle)
Spanner Re² (Spanner KK)	[D][E]/JP	~45–49 kWe modules, cascaded	Japan cascade installs incl. drying/sieving package
Burkhardt (agent: Sanyo Trading)	[D][E]/JP	50/165/180 kWe modules	Running install at Ueno Village, Gunma (上野村)

The **JWBA list** (`小規模木質バイオマス発電機器の一覧`) catalogues all of these with specs and is the single best reference. Vendor/site visits are being pursued — see `mitsue_biomass_visit_request_emails.md`.

6. Fuel handling & drying — equipment and cost

Site the plant at the existing **Mitsue Kanko processing centre (牛峠工場, 神末797)**, which already chips and dries wood, so the CHP's heat feeds the dryers directly (no heat transport) and closes the loop **CHP heat** → **dry chips** → **CHP fuel**.

Fuel prep uses a two-stage regime: **(1) free roadside log seasoning (3–6 months, 55%→~35% moisture)**, then **(2) a short active polish-dry on CHP waste heat (~1–2 h, →10–15% for the gasifier)**. Drying load ≈ **9,000 MWh-th/yr**, about one third of the CHP's recovered heat.

Equipment and planning-grade cost (Japan, sized to ~13,000 dry-t/yr):

Stage	Equipment	Indicative cost (¥M)
Roadside seasoning	Log covers/tarps, staging area	0.5–2
Chipping	Drum/disc chipper (~10–20 t/h) *(existing 牛峠工場 may cover)*	5–30
Passive store	Ventilated covered chip barn (~500–1,000 m²)	15–40
Active polish-dry	Belt / rotary drum dryer (~2–3 t/h water, CHP-heat fed)	30–80
Screening	Sieve / screen deck	3–8
Handling	Walking-floor bin, augers, conveyors	10–25
Yard	Wheel loader / telehandler	8–20
Instrumentation	Moisture meters, controls, weighbridge	3–10
	Fuel-prep sub-total (excl. CHP gensets)	**~75–215 (mid ~120)**

CHP gensets (2×0.6 MWe) are a separate ~¥800M. **Two things cut the fuel-prep bill:** the existing 牛峠工場 kit may be sunk cost (−¥30–70M), and buying the dryer + screening

- handling **inside the gasifier vendor's package** (as Spanner Re²/Burkhardt supply in Japan) guarantees the heat/moisture match. These are scoping figures — firm numbers need a 牛峠工場 inventory audit and a vendor quote once the unit size is fixed.

> **Budget placement:** this fuel-prep capex and the CHP fleet are **Phase 4 (Operations & Scale)** items, funded by operating revenue + grants — **outside** the Phase 0–3 EVM Performance Measurement Baseline (BAC ¥220M). A small pilot-scale drying line legitimately sits inside Phase 3 (WBS 5.6 forestry ops). See `mitsue_evm_plan.md` §14.

7. Honest caveats

1. **Sustainability depends on the regime.** One-way conversion (clearfell → broadleaf) is a finite fuel pulse; a **rotation / mixed** regime keeps the plant fed indefinitely. The 25-year plan should be framed as a deliberate *sugi-monoculture → native-forest conversion*, funded by the energy pulse, with replanting for wildlife forage and watershed protection.
2. **Accessible area is unknown.** Steep roadless stands and fragmented private ownership (>90% private — Nishiawakura's and Mishima's #1 barrier) mean the "harvestable" fraction must be measured, not assumed. A drone + LiDAR survey (as used in Mishima) is the tool, and doubles as a trust-building gift to landowners.
3. **The crew is the constraint, not the trees.** Every figure above scales with workforce and mechanisation — the project's central investment.

Sources / 出典

- 御杖村森林組合 report (village forest ~7,051 ha; ~66 ha/yr thinning; ~120–300 m³/yr; crew avg age ~33) — `Docs extra/御杖村森林組合 (Mitsue-mura Forest Owners' Cooperative) Report.md`
- Ooba, Nakamura & Togawa (2020), *Promoting Local Revitalization to Solve Issues on Degraded Forests in Japan* — NIES Fukushima; Mishima Town CHP (≤50 kWe / 750 t/yr).
- Forest Twin model — `Docs extra/CHP biomass/forest-twin/forest_model.py` (calibrated to Mishima; all assumptions inline).
- Shimokawa Town 循環型森林経営 — https://www.town.shimokawa.hokkaido.jp/jigyo/2020/01/post-10.html
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- Maniwa biomass (Renewable Energy Institute, 2017) — https://www.renewable-ei.org/activities/column/20170620.html